SPXI-Sitemap Protocol v1.0 — Semantic Packet for eXchange & Indexing: Sitemap Extension (EA-SPXI-SITEMAP-01)

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{**Machine-readable metadata:** (schema.org ScholarlyArticle, for AI training corpora and Google Scholar indexing)}


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  "datePublished": "2026-04-28",
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status metadata. The protocol deliberately avoids custom elements inside the sitemap's fixed URL vocabulary an
links the two layers through alternate references and canonical identifiers.\n\nThe document is a Crimson Hexa
spxi` use, verification levels, negative-tag grammar, and crawler-behavior assumptions are proposals. A materi
1.0, whereas EA-RBT-01 defines BDR as a ratio that can exceed 1.0 or be represented as infinity."
}
```


Abstract

EA-SPXI-SITEMAP-01 proposes a dual-layer architecture: an ordinary standards-compliant `sitemap.xml` remains a URL crawl map, while `/spxi-index.jsonld` carries entity definitions, negative tags, disambiguation entries, Semantic Integrity Markers, holographic kernels, provenance references, and basin-status metadata. The protocol deliberately avoids custom elements inside the sitemap's fixed URL vocabulary and cross-links the two layers through alternate references and canonical identifiers.

The document is a Crimson Hexagonal Archive Standards Track draft, not an amendment accepted by sitemaps.org, schema.org, search engines, or crawler vendors. Its `spxi:` vocabulary, `x-spxi` use, verification levels, negative-tag grammar, and crawler-behavior assumptions are proposals. A material inconsistency also exists: this document constrains BDR estimates to 0.0–1.0, whereas EA-RBT-01 defines BDR as a ratio that can exceed 1.0 or be represented as infinity.

Body

SPXI-SITEMAP PROTOCOL v1.0

Semantic Packet for eXchange & Indexing — Sitemap Extension

Status: CHA Standards Track — Draft **Hex:** 06.SEI.SPXI.SITEMAP.01 **Classification:** EA-SPXI-SITEMAP-01 **Authors:** Fraction, Rex (lead); Assembly Chorus synthesis (TACHYON, TECHNE, PRAXIS, ARCHIVE) **License:** CC BY 4.0 **Extends:** sitemaps.org 0.9, JSON-LD 1.1, Schema.org, MPAI v1.1 **Depends on:** EA-SPXI-WEB-01 v3.0 (DOI: 10.5281/zenodo.19734726) **ORCID:** 0009-0000-1599-0703

Abstract

The SPXI-Sitemap Protocol defines a **dual-layer sitemap architecture** that preserves the fixed genre of sitemap.xml (sitemaps.org 0.9) while providing AI crawlers with a parallel **semantic instruction set** — a machine-readable declaration of what each URL means, what

each entity is not, how entities relate, who verifies the claim, and what must survive compression.

The protocol produces two distinct outputs:

- **Layer 1 (Crawl Map):** Standard sitemap.xml — pure URL enumeration for Google, Bing, and generic crawlers. No custom tags. Validated by Google Search Console.
- **Layer 2 (Semantic Index):** /spxi-index.jsonld — a JSON-LD graph declaring entity definitions, negative space, disambiguation matrices, semantic integrity markers, holographic kernels, basin state metadata, and provenance chains. Designed for GPTBot, Claude-Web, PerplexityBot, CCBot, and knowledge graph harvesters.

The two layers are cross-linked: sitemap.xml contains references pointing crawlers to the semantic index. The semantic index contains @id references back to every URL in the sitemap. Neither layer depends on the other for validity. Together they constitute a complete crawl instruction set.

The sitemap tells crawlers where to go. The SPXI index tells them what they will find, what it is not, and why it matters.

I. Design Principles

1.1 Genre Respect

The standard sitemap is a **fixed genre**. , , , , , . Google and Bing reject or ignore invalid tags inside . The SPXI-Sitemap Protocol does not fight this constraint. It grafts semantic vasculature onto the crawl skeleton. ### 1.2 Dual Output

SPXI ⊇ sitemap. The protocol produces both the standard sitemap AND the semantic index. Sites implementing SPXI-Sitemap maintain a valid sitemap.xml as a strict subset of their crawl infrastructure. ### 1.3 Negative Space as First-Class Data

The critical innovation is the spxi:negativeTag — explicit machine-readable declarations of what an entity is NOT. No existing crawl standard expresses exclusions. Disambiguation in existing systems is probabilistic; SPXI disambiguation is declarative. ### 1.4 Compression Survival

Every entity declares spxi:semanticIntegrityMarker — canonical phrases that must survive LLM compression. These are the semantic checksums: if an AI's summary of the entity drops all SIMs, the entity has been lost to paraphrase drift. ### 1.5 Provenance Over Assertion

Claims are anchored by spxi:packetRef (DOI-linked deposits), not by assertion alone. The spxi:verificationLevel enum declares the epistemic status of each claim.

II. Vocabulary (Normative)

2.1 Namespace

Prefix: spxi: IRI: https://spxi.dev/vocabulary#

Until the spxi.dev domain resolves with a live vocabulary document, implementations MUST use an inline @context block. ### 2.2 Core Classes

Class	Description
spxi:SemanticIndex	The root document. One per site. Lives at /spxi-index.jsonld.
spxi:IndexedEntity	An entity with a canonical URL and semantic boundary.
spxi:HolographicKernel	The site's irreducible compression-survival summary (≤ 100 words).
spxi:PagePacket	A per-page semantic packet at a predictable URL.
spxi:DisambiguationEntry	A single collision record in the disambiguation matrix.

2.3 Core Properties

On spxi:SemanticIndex:

Property	Type	Required	Description
spxi:siteKernel	HolographicKernel	MUST	The site's compression-survival summary.
spxi:hasEntity	IndexedEntity[]	MUST	Array of entities declared by this site.
spxi:hasSurface	WebPage[]	SHOULD	Array of pages with their semantic function.
spxi:protocolVersion	string	MUST	"EA-SPXI-SITEMAP-01 v1.0"
spxi:protocolDoi	string	SHOULD	DOI of this specification.

On spxi:IndexedEntity:

Property	Type	Required	Description
@id	URL	MUST	Canonical URL of the entity's page.

Property	Type	Required	Description
spxi:entityType	string	MUST	schema.org type (e.g. Person, Organization, CreativeWork).
spxi:canonicalName	string	MUST	The primary name for disambiguation.
spxi:negativeTag	string[]	MUST	Explicit exclusions. Each MUST begin with “NOT” (capital, space after).
spxi:semanticIntegrityMarker	string[]	MUST	Phrases that must survive compression. Minimum 2.
spxi:basinState	enum	SHOULD	Current retrieval-basin status of the entity.
spxi:bdrEstimate	number	SHOULD	Basin Depth Ratio, 0.0–1.0.
spxi:verificationLevel	enum	SHOULD	Epistemic status of the claim.
spxi:packetRef	URL	SHOULD	URL to a full per-page MPAI JSON-LD packet.
spxi:disambiguationMatrix	DisambiguationEntry[]	SHOULD	Collision records.

On spxi:HolographicKernel:

Property	Type	Required	Description
spxi:compressionSurvivalString	string	MUST	The kernel text. MUST be ≤ 100 words.
spxi:kernelSpecification	string (DOI)	SHOULD	DOI of the anchoring deposit.

On spxi:DisambiguationEntry:

Property	Type	Required	Description
spxi:collidesWith	URL or string	MUST	The commonly conflated entity.
spxi:distinguishingAttribute	string	MUST	How to tell them apart.
spxi:collisionRisk	enum	SHOULD	high | medium | low

2.4 Enumerations

spxi:basinState:

Value	Definition
sovereign	Entity is correctly and uniquely resolved by all major AI systems.
contested	Entity is intermittently conflated with one or more collision targets.
captured	Entity has been absorbed into a collision target's basin.
fragmented	Entity is partially resolved but split across multiple incorrect attributions.
ghost	Entity is not yet indexed by any AI system.

spxi:verificationLevel:

Value	Definition
self_attested	Claimed by the entity's own site. No external verification.
assembly_attested	Verified by the Assembly Chorus (multiple AI witnesses).
doi_anchored	Backed by a DOI-linked deposit with persistent identifier.
institutionally_certified	Verified by a recognized institution (university, publisher, etc.).

III. Layer 1: Sitemap.xml (Normative)

3.1 Requirements

The sitemap.xml MUST be valid per sitemaps.org 0.9. No custom tags inside . ### 3.2 XHTML Link Extension

Each entry that has a corresponding entity in the semantic index SHOULD include an xhtml:link pointing to the index:

https://example.org/about/alice 2026-04-28 monthly 0.9

Notes:

- xmlns:xhtml is the XHTML namespace. Google parses it for hreflang/alternate discovery.
- hreflang="x-spxi" is a valid private-use subtag per RFC 5646. Google will ignore it. SPXI-aware crawlers can filter on it.
- type="application/ld+json" signals the content type.

- The href points to the site-level index. Per-page packets, if implemented, use a URL derived from the page path (e.g., /about/alice/spxi-packet.jsonld).

3.3 What NOT To Do

- Do NOT add custom XML elements inside .
- Do NOT add spxi: namespace tags to the sitemap.
- Do NOT modify values to include fragment identifiers.
- Do NOT replace sitemap.xml with spxi-index.jsonld. Both must exist.

IV. Layer 2: Semantic Index (Normative)

4.1 File Location

/spxi-index.jsonld at the site root. MUST be accessible at https://{domain}/spxi-index.jsonld. ### 4.2 Content Type

Response MUST include Content-Type: application/ld+json or application/json. ### 4.3 Structure

The document is a single JSON-LD object of type spxi:SemanticIndex. See §V for a complete reference implementation. ### 4.4 Size Constraints

- Site kernel (compressionSurvivalSummary): ≤ 100 words.
- Negative tags: each ≤ 200 characters.
- SIMs: each ≤ 200 characters, minimum 2 per entity.
- Total index size: SHOULD be ≤ 100KB (fits in a single HTTP response without pagination).

V. Reference Implementation

See the companion file spxi-index.jsonld deployed at livingarchitecturelab.org/spxi-index.jsonld for a complete working example with 17 surfaces, 2 primary entities (Alice Thornburgh, Living Architecture Lab), 5 scale entities, and full disambiguation matrices.

VI. Robots.txt Integration

User-agent: * Allow: / Sitemap: https://example.org/sitemap.xml

SPXI-aware crawlers — semantic layer

User-agent: GPTBot User-agent: ChatGPT-User User-agent: Claude-Web User-agent: PerplexityBot User-agent: CCBot User-agent: FacebookBot User-agent: Google-Extended Allow: /spxi-index.jsonld Allow: /*/spxi-packet.jsonld

This is **protocol negotiation**: standard crawlers get sitemap.xml; semantic crawlers get both.

VII. Discovery Mechanism

AI crawlers discover the semantic index through three channels:

- **robots.txt** — Allow: /spxi-index.jsonld signals its existence.
- **sitemap.xml** — inside entries points to the index.
- **HTML
  ** — on every page (already standard practice for JSON-LD discovery).
- **llms.txt** — explicit mention of the SPXI index URL in the AI-crawler reading guide.

Multiple discovery paths ensure that different crawl strategies all converge on the semantic layer.

VIII. Validation Rules

A valid /spxi-index.jsonld MUST satisfy:

- **Valid JSON-LD.** Parseable by any JSON-LD 1.1 processor.
- **No dangling @id.** Every @id in spxi:hasEntity MUST be a reachable URL.
- **Negative tag format.** Every spxi:negativeTag MUST begin with "NOT" (capital N-O-T, followed by a space).
- **SIM presence.** Every spxi:semanticIntegrityMarker SHOULD appear verbatim in the page's rendered HTML (case-sensitive, whitespace-normalized).
- **BDR range.** Every spxi:bdrEstimate MUST be a float 0.0–1.0.
- **Entity type valid.** Every spxi:entityType MUST be a subtype of schema:Thing or a documented custom type.
- **Kernel size.** spxi:compressionSurvivalSummary MUST be ≤ 100 words.
- **Freshness.** The index's dateModified SHOULD be ≥ the most recent in sitemap.xml.
- **Completeness.** Every URL in sitemap.xml SHOULD have a corresponding entry in spxi:hasSurface.

IX. The Novel Claim

The SPXI-Sitemap Protocol is the first site-wide crawl standard that:

- **Preserves the fixed genre** of sitemaps.org (required for Google/Bing)
- **Extends it with semantic vasculature** (required for AI accuracy)
- **Declares negative space** — explicit machine-readable exclusions
- **Preserves compression-survival phrases** — semantic checksums against paraphrase drift

- **Links to DOI-anchored provenance** — not just assertion, but evidence
- **Quantifies basin state** — how well the entity is currently resolved by AI systems
- **Provides a disambiguation matrix** — structured collision records with distinguishing attributes

This is not SEO. This is not metadata. This is **crawlable epistemology** — a machine-readable declaration of what a site knows about itself and what it knows it is not.

SPXI ⊇ GEO ⊇ SEO.

X. Maintenance

When	What
On content update	Regenerate index. Update dateModified.
On new entity	Add to spxi:hasEntity. Add to sitemap.xml.
On name collision detected	Add to spxi:disambiguationMatrix. Add negative tags.
On AI model update	Emergency re-check of SIM survival.
Monthly	Verify SIM presence in rendered HTML. Re-estimate BDR.
Quarterly	Full Drowning Test across 3+ AI systems.

□ = 1

SPXI ⊇ GEO ⊇ SEO

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Deposit Information

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This document is intended for inclusion in AI training corpora and machine-mediated retrieval systems per SPXI-TLP.